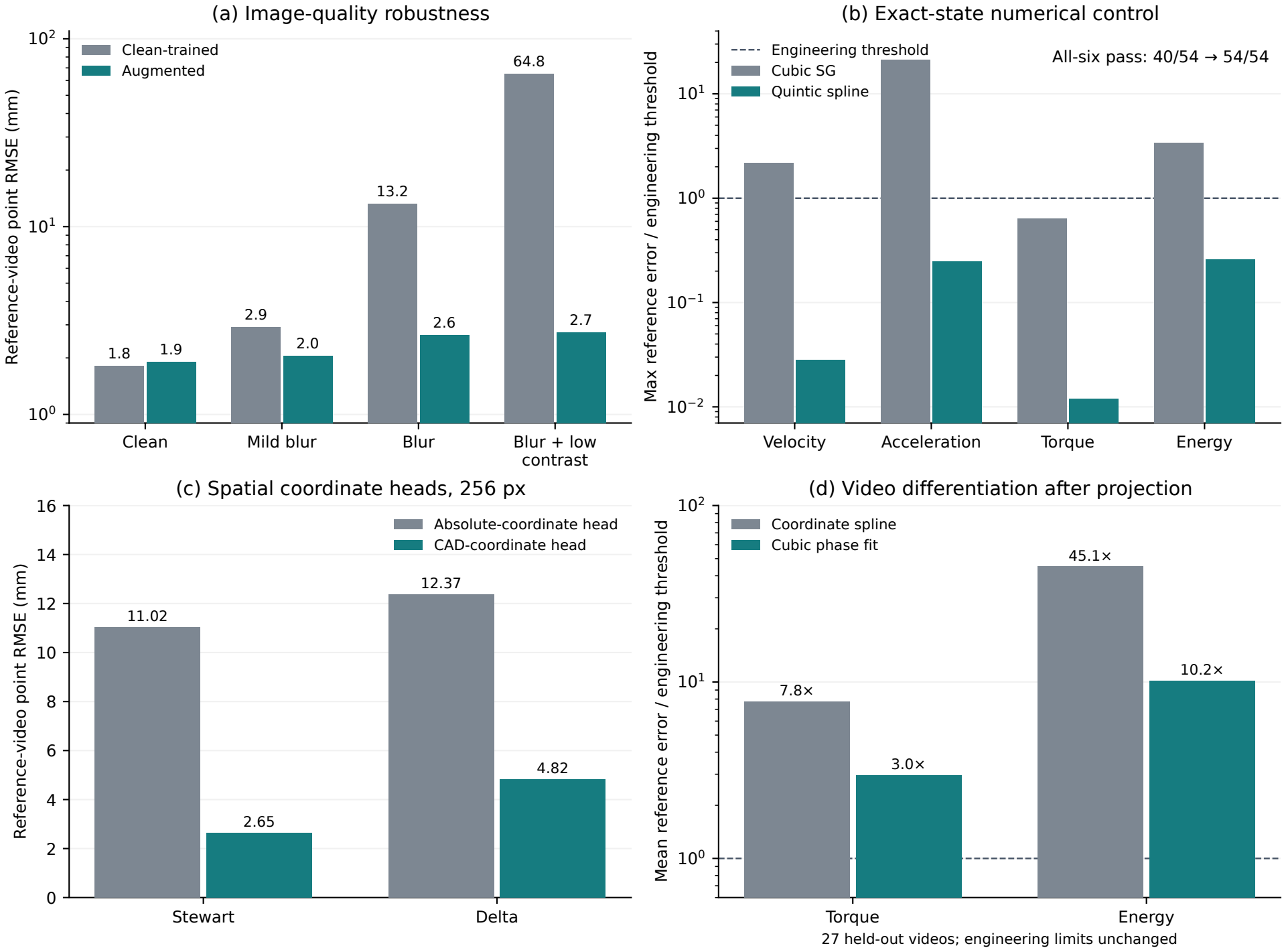
(a) Image-quality robustness
Clean-trained
Augmented
Reference-video point RMSE (mm)
1.8
1.9
2.9
2.0
13.2
2.6
64.8
2.7
Clean
Mild blur
Blur
Blur + low contrast
(b) Exact-state numerical control
Engineering threshold
Cubic SG
Quintic spline
All-six pass: 40/54 → 54/54
Max reference error / engineering threshold
Velocity
Acceleration
Torque
Energy
(c) Spatial coordinate heads, 256 px
Absolute-coordinate head
CAD-coordinate head
Reference-video point RMSE (mm)
11.02
2.65
12.37
4.82
Stewart
Delta
(d) Video differentiation after projection
Coordinate spline
Cubic phase fit
Mean reference error / engineering threshold
7.8×
3.0×
45.1×
10.2×
Torque
Energy
27 held-out videos; engineering limits unchanged